Trust the View That Sees the Target: Mining Cross-View Conflicts for Reliability-Gated Disaster Damage Assessment

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Abstract

After a disaster, building damage is assessed from overhead tiles and ground-level photographs, and most methods fuse the two views symmetrically. This paper studies the samples where the views do not agree: the *conflict cases*, on which two independently trained single-view models disagree. We mine them from three paired collections (inspection photographs from the 2025 Eaton wildfire and street-view panoramas from Hurricanes Ian and Milton, each matched to very-high-resolution overhead tiles), where they make up 9.6–33.2% of the test data. An oracle that picks the correct single view on each conflict exceeds the best evaluated fusion method by 0.19–0.32 conflict accuracy. A linear reliability gate over building-visibility features, calibrated confidences, and disagreement recovers part of this headroom where ground photographs are aimed at the building: on Eaton it beats end-to-end fusion by +0.072 conflict accuracy ($p < 10^{-4}$) and calibrated averaging by +0.051 ($p = 0.0001$). On the panoramic datasets it is statistically indistinguishable from end-to-end fusion. Cropping panoramas toward the building raises the conflict gain of fusion, whereas random crops of the same size do not. Finally, the spatial density of conflicts, computed without labels, correlates with tile-level wildfire damage (Spearman $r = 0.615$, 25 tiles). Mining conflicts turns "does fusion help?" into "which view should be trusted, where, and why?".

CCS Concepts

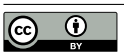
• **Computing methodologies** → **Computer vision**; *Machine learning*; • **Information systems** → *Geographic information systems.*

Keywords

cross-view fusion, street-view imagery, satellite imagery, disaster damage assessment, reliability estimation, GeoAI

ACM Reference Format:
Yifan Yang and Lei Zou. 2026. Trust the View That Sees the Target: Mining Cross-View Conflicts for Reliability-Gated Disaster Damage Assessment. In *The 5th ACM SIGSPATIAL International Workshop on Searching and Mining Large Collections of Geospatial Data (GeoSearch '26), November 03–06, 2026, Riverside, CA, USA.* ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3849732.3857333

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GeoSearch '26, Riverside, CA, USA
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ACM ISBN 979-8-4007-3058-0/2026/11
https://doi.org/10.1145/3849732.3857333

1 Introduction

Building-level damage assessment after a disaster increasingly draws on two large image collections at once. Overhead imagery covers a whole region within hours and is the basis of standardized damage mapping [2]. Ground-level imagery, whether crowdsourced street-view panoramas or official inspection photographs, shows the facades and debris that overhead sensors cannot see [11, 13]. Methods that pair the two views serve geolocalization and damage mapping [5, 6, 12], multimodal arbitration [10], and agentic assessment pipelines [8, 9]. Their usual conclusion is encouraging: fusion improves average accuracy over either view alone.

Average accuracy hides both the value and the limitation of cross-view fusion: for most buildings the two views agree. The informative records are the *conflicts*, the samples on which independently trained single-view models disagree. A conflict often means that one sensor cannot see the damage: a facade can look intact from above and gutted from the street, or trees can hide a destroyed roof from the road. Finding these records, deciding which view to believe for each, and reading their spatial pattern is a search-and-mining problem as much as a modeling one.

We therefore make conflict cases the unit of analysis and ask four questions. (1) How much headroom do fusion methods leave on conflicts? We compare them with an oracle that trusts whichever single view is correct. (2) Can a simple, readable gate recover part of it from cues about whether the ground image actually shows the building? (3) Does aiming the ground view at the building change the value of fusion? We vary the framing *within* a dataset with a controlled cropping experiment. (4) Does the spatial density of conflicts, which needs no labels once the models are trained, track damage (Figure 1)? Code, protocols, and derived result tables: https://github.com/rayford295/CrossViewGate.

2 Data and Protocol

We use three paired street/overhead collections spanning two hazards and, more importantly, two ground-view *observation regimes*: property-centric photographs aimed at a building, and panoramas that are not.

Eaton wildfire (Altadena, CA; property-centric). Building-level CAL FIRE Damage Inspection (DINS) photographs taken after the January 2025 fire, paired by location with post-event VHR overhead tiles. The six-level DINS scale is mapped to three ordinal classes (no/trace damage, damaged–repairable, destroyed). The split has 6.5k/1,966/1,984 train/val/test pairs. **Hurricane Ian**

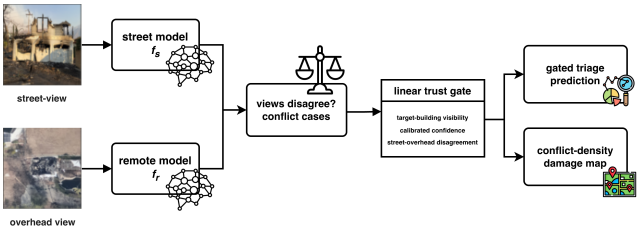


Figure 1: Overview. Single-view models define the conflict cases; a linear gate decides how much to trust each predictor per sample (the implemented gate also mixes in the end-to-end fusion model, omitted here); conflict density is a label-free damage indicator. Shown: an Eaton-fire conflict.

Table 1: Test-set results (five-seed means; accuracy std ≤ 0.03). Conflict accuracy is computed on each seed's conflict subset. Closure follows Eq. (1) applied to the displayed means; the oracle reaches 0.961 / 0.896 / 0.964 conflict accuracy, i.e. 0.475 / 0.367 / 0.407 above the better single view.

	Eaton wildfire (conflict rate 9.6%)			Hurricane Ian (33.2%)			Hurricane Milton (25.7%)		
Method	Acc.	Conflict acc.	Closure	Acc.	Conflict acc.	Closure	Acc.	Conflict acc.	Closure
street_only	0.931	0.486	—	0.703	0.529	—	0.759	0.557	—
remote_only	0.929	0.475	—	0.648	0.367	—	0.721	0.407	—
concat	0.930	0.674	0.40	0.709	0.588	0.16	0.766	0.622	0.16
crossview (end-to-end fusion)	0.940	0.699	0.45	**0.735**	**0.624**	**0.26**	0.776	**0.649**	**0.23**
calibrated probability averaging	0.954	0.732	0.52	0.731	0.617	0.24	0.776	0.626	0.17
reliability gate (linear)	**0.959**	**0.768**	**0.59**	0.731	0.618	0.24	**0.777**	0.631	0.18

(panoramic). The CVIAN pairing released with CVDisaster [6]: Mapillary 360° panoramas over Sanibel Island, 27 cm post-event satellite imagery, and three-level damage-perception labels, with 3,248/573/300 pairs. **Hurricane Milton (panoramic).** A 360° street-view/satellite pairing over Florida after the October 2024 hurricane, built in the same way; 1,740/307/254 pairs.

Splits and leakage. Eaton is split by property and tile, so all photographs of one structure fall in one partition. For Ian the headline numbers keep the released split for comparability with CVDisaster. However, an audit against the official CVIAN position file found that 299 of the 300 test samples share a Mapillary sequence with the training data, so we also evaluate a repaired protocol with 0.005° spatial blocks and a 25 m buffer. Under it, macro-F1 drops by 0.05–0.10 for every model, so index-based splits overstate spatial generalization, although the single-view ranking (street above satellite) is unchanged. The Ian results in Table 1 should therefore be read as in-distribution results, not as evidence of spatial transfer.

3 Method

Base models and conflict cases. For each dataset we train four models under one protocol: *street_only* and *remote_only* (ResNet-18 [3] with a linear head), *concat* (both encoders with an MLP head), and *crossview* (both encoders with a learned interaction head), which serves as the end-to-end fusion reference. Training uses class-balanced cross-entropy, 224-pixel inputs, AdamW, up to 15 epochs with early stopping on validation macro-F1, and five seeds. Writing $s(x)$ and $r(x)$ for the two single-view predictions, the **conflict subset** is $C = \{x : s(x) \neq r(x)\}$. Conflict rates are 9.6% on Eaton, 33.2% on Ian, and 25.7% on Milton.

Oracle single-view gap. The oracle counts a conflict sample as correct if *either* single view is correct. For a method m we report its conflict accuracy $\mathrm{acc}_C(m)$ and its **oracle-gap closure**,

$$\mathrm{closure}(m) = \frac{\mathrm{acc}_C(m) - \mathrm{acc}_C(\text{best single})}{\mathrm{acc}_C(\text{oracle}) - \mathrm{acc}_C(\text{best single})}, \tag{1}$$

where "best single" is the single-view model with the higher mean conflict accuracy. The oracle only chooses *which* of two fixed models to trust, so it measures the headroom of hard view selection; it is not an upper bound for probability mixtures, which can in principle exceed it.

Calibration. Since miscalibration can distort averaging [1], we fit one temperature per view on validation predictions and evaluate every training-free fusion baseline on calibrated probabilities; the argmax, and hence the conflict subset, is unchanged.

Visibility-conditioned reliability gate. A frozen SegFormer-B0 [7] trained on ADE20K gives four building-visibility features per street image: the building pixel ratio, the same ratio inside a centered crop of half the image size, their difference (a centering signal), and the normalized distance of the building centroid from the image center. These features describe visible built structures in general, not the specific target building. Seven more features describe the predictions themselves: calibrated confidence and entropy for each view, the confidence gap, the Jensen–Shannon divergence between the two calibrated distributions, and a disagreement flag. Given the eleven z-scored features z, the gate is a mixture of experts [4] with linear logits,

$$p(y \mid x) = \sum_k w_k(z)\, p_k(y \mid x), \qquad w(z) = \mathrm{softmax}(Wz + b), \tag{2}$$

over $k \in \{\text{street}, \text{remote}, \text{crossview}\}$, fit on the validation split by minimizing the negative log-likelihood of the mixture. A *two-expert* variant (street and remote only) has a single logit, so each coefficient

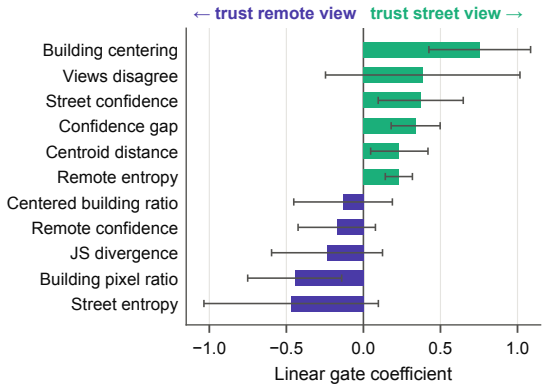


Figure 2: Coefficients of the two-expert (street vs. remote) linear gate on Eaton (mean ± std over five seeds). Positive values shift trust toward the street view. Bars whose error bars cross zero do not have a stable sign across seeds.

reads directly as a shift of trust toward one view. A small MLP in place of the linear logits did not help on Eaton.

Field-of-view intervention. Each 1024 × 512 panorama is cropped to a 256 × 256 window with a 90° horizontal field of view on the horizon band. In the **building-centered** variant the window is aimed at the circular-mean azimuth of the SegFormer building mask. In the **random** variant it is aimed at a reproducible uniformly random azimuth. The two variants share output size, projection, and resolution and differ only in where the window points. We retrain *street_only* and *crossview* on each variant (three seeds, shorter three-epoch schedule; the original-panorama reference uses the same schedule), reuse the remote models, and compare the gain of fusion on conflicts.

Conflict density. We average test-set predictions over seeds, aggregate them to geographic tiles, and compute per tile the hard conflict density (fraction of samples on which the two argmax predictions disagree) and, as a control, the mean single-view predictive entropy. Both are rank-correlated (Spearman) with the tile's mean ordinal damage label.

4 Results

Overall results and the oracle gap. For paired comparisons we average each test sample's correctness difference over the seeds in which it is a conflict and apply a two-sided sign-flip permutation test (20,000 permutations). These pooled differences weight samples rather than seeds, so they differ slightly from the differences of the means in Table 1. First, end-to-end fusion beats the street model on conflicts on every dataset (Eaton +0.246, $p < 10^{-4}$; Ian +0.078, $p = 0.013$; Milton +0.141, $p = 0.006$). Second, once the single-view models are converged and calibrated, probability averaging is a strong baseline: on Eaton it exceeds end-to-end fusion in both overall accuracy (0.954 vs. 0.940) and conflict accuracy (0.732 vs. 0.699). Third, the oracle headroom dwarfs these differences: picking the right single view per sample would add 0.475 / 0.367 / 0.407 conflict accuracy over the better single view, and still 0.193 / 0.272 / 0.315 over the best evaluated fusion method. The unused information concerns *when* each view is reliable.

Table 2: Field-of-view intervention (three seeds, three-epoch schedule; remote models unchanged). Conflict gain = crossview − best single view on each variant's own conflict subset.

Dataset	Street-view variant	Street acc.	Conflict gain
Ian	original panorama	0.709	+0.064
Ian	building-centered crop	0.648	**+0.109**
Ian	random crop (control)	0.574	+0.042
Milton	original panorama	0.740	+0.067
Milton	building-centered crop	0.693	**+0.149**
Milton	random crop (control)	0.654	+0.055

The reliability gate. On the property-centric dataset the gate gains +0.072 conflict accuracy over end-to-end fusion ($p < 10^{-4}$) and +0.018 on the full test set ($p < 10^{-4}$); against calibrated averaging, the strongest baseline, it gains +0.051 on conflicts (95% bootstrap CI [+0.026, +0.077], $p = 0.0001$). Its closure of 0.59 is the highest in Table 1. On the panoramic datasets the street views rarely show a building prominently (mean building pixel ratio 0.027 on Ian vs. 0.119 on Eaton), so the visibility features carry little signal, and the gate is statistically indistinguishable from end-to-end fusion (pooled differences +0.005, $p = 0.70$ on Ian; −0.015, $p = 0.57$ on Milton). How much of the Eaton gain comes from visibility-based weighting rather than from adding a third expert? The two-expert gate, which can only reweight street and remote, reaches 0.734 conflict accuracy, on par with calibrated averaging (0.732) and above end-to-end fusion (+0.034, $p = 0.047$). Adding crossview as a third expert lifts this to 0.768, so the headline gain combines both effects. The two-expert coefficients (Figure 2) are the interpretable part. The largest weight is building centering (+0.75 ± 0.33), followed by the confidence gap and street confidence, while the raw building pixel ratio is negative (−0.45 ± 0.31). Trust in the street view thus rises when the photograph is confident and the building sits in the middle of the frame, not merely when much of it is built-up area. Street entropy and the disagreement flag have large but seed-dependent weights.

Field-of-view intervention. Building-centered cropping raises the conflict gain of fusion on both hurricanes (Table 2): from 0.064 to 0.109 on Ian and from 0.067 to 0.149 on Milton. Random crops with identical geometry do not (0.042 and 0.055), and the building-minus-random contrast is positive in all six dataset–seed pairs. Cropping *lowers* street-only accuracy yet *raises* the benefit of fusion, so the gain does not come from a stronger street model, and the random control rules out cropping artifacts. The result supports building-oriented framing as a driver of the regime effect; since the segmenter does not identify the target building and retraining changes the conflict subsets, a fixed-sample test with target annotations would be stronger.

Conflict density as a spatial damage signal. On the wildfire, the conflict density of an overhead tile tracks its damage (Spearman $r = 0.615$, $p = 0.001$ over 25 tiles; $r = 0.60$ on the 0.01° grid of Figure 3). The uncertainty control behaves very differently: it *anti-correlates* with damage ($r = -0.37$ to -0.52), because parcels burned to the ground are easy, confident classifications. Conflict density is thus not model uncertainty under another name. Damage on Eaton is spatially autocorrelated (Moran's $I = 0.27$), so the nominal

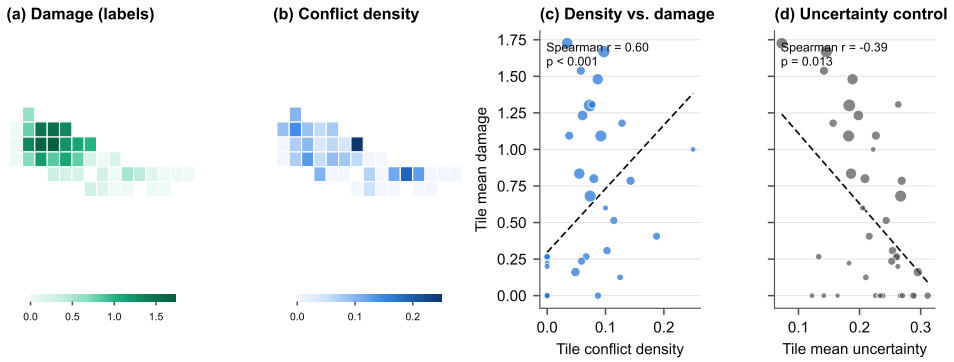


Figure 3: Conflict density as a label-free damage indicator (Eaton wildfire, 0.01° grid). (a) Labeled tile damage. (b) Tile conflict density, computed from model predictions without labels. (c) Their rank correlation. (d) The single-view uncertainty control, which anti-correlates with damage. Point size reflects tile sample count; p-values are nominal and ignore spatial dependence.

p-values overstate the evidence. On Milton the signal is weaker and not significant once the models converge ($r = 0.289$, $p = 0.14$), because better single-view models disagree less.

5 Discussion and Conclusion

The oracle headroom, a restricted choice between two existing models, is larger than any fusion gain we observed, so for paired geospatial collections the productive question is less "how should features be fused?" than "when should each view be believed?". The gate adds one frozen segmentation pass per street image and a linear fit on existing probabilities, and its two-expert form reads as one sentence (*trust the street view when it is confident and the building is centered*), which matters where automated triage must be auditable. For panoramic imagery, calibrated averaging or end-to-end fusion suffices; for property-centric photographs, as in inspection programs, gating pays. Capture policy is also a lever: framing ground photographs on structures raised the value of fusion.

Limitations. The property-centric regime rests on one wildfire dataset; Milton is small (254 test pairs); the released Ian split has sequence overlap and the repaired protocol only six test blocks. The permutation tests treat test samples as independent, ignoring dependence between nearby properties. The density signal relies on supervised models and needs spatially aware inference and replication on further disasters.

Conclusion. Where the two views disagree, symmetric fusion leaves substantial accuracy unused. A visibility-conditioned linear gate recovers part of it when ground photographs are aimed at the building, and spatially aggregated disagreement is a label-free damage indicator. Mining conflicts turns "does fusion help?" into "which view should be trusted, where, and why?".

Acknowledgments

Claude (Anthropic) and OpenAI Codex were used for language editing and for checking reported numbers against the result logs. The authors take full responsibility for the content.

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